4th Multi-modal Aerial View Image Challenge - Translation: Our Solution

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Figure 1. Teaser: Our proposed solution for multimodal aerial-view image translation across EO, IR, and SAR modalities.

Abstract

In this paper, we present a comprehensive study on cross-modal remote sensing image-to-image translation.

1. Introduction

The Multi-modal Aerial View Image Challenge for cross domain image translation [1–3] is a leading challenge held in conjunction with the Perception Beyond the Visible Spectrum (PBVS) workshop from 2023 to 2025. Data collected across multiple sensing modalities—such as Electro-Optical (EO), Infrared (IR), and Synthetic Aperture Radar (SAR)—each capture distinct characteristics and measurements of objects, providing complementary perspectives on the same scene. However, data coverage limitations and modality-specific constraints often hinder the full utilization of these diverse sources. For instance, while SAR sensors offer all-weather capabilities and can penetrate atmospheric obstructions, their imagery is complex to interpret and publicly available datasets are scarce. Conversely, widely available EO imagery is limited by lighting conditions and atmospheric visibility. Domain translation seeks to bridge this gap by converting one modality into another, enabling cross-modal data synthesis and enhancing sensor interoperability. By focusing on translation tasks such as RGB→IR, SAR→EO, SAR→IR, and SAR→RGB, the MAVIC-T challenge aims to bolster existing data volume, increase data diversity across regions, and serve as a platform for advancing conditioned image synthesis techniques, ultimately fostering the development of more adaptable and robust models for multi-modal image analysis.

interpret SAR need special knowledge [4].

In this paper, we propose [YourMethodName], a novel [Diffusion/GAN/Transformer] framework designed to [solve specific problem]. Our approach incorporates a [Specific Module, e.g., 'Latent Consistency Constraint'] and a [Specific Loss] to ensure that the translated images maintain the precise geometric layout of the source modality while achieving high photorealism.

2. Related Work

Image translation methods in general domain across diverse sensor modalities, include grayscale, near-infrared (NIR), thermal image colorization, and color transfer functions [4–7].

3. Preliminary

Problem Formulation Let $\mathcal{X} \subset \mathbb{R}^{C_s \times H \times W}$ denote the source domain and $\mathcal{Y} \subset \mathbb{R}^{C_t \times H \times W}$ denote the target domain, where C_s and C_t are the number of channels in the source and target modalities, respectively. Given a set of spatially-aligned training pairs $\{(x_i, y_i)\}_{i=1}^{N}$ with $x_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$, the goal of cross-modal image-to-image translation is to learn a mapping $G : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes a composite distance between the generated image $\hat{y} = G(x)$ and the ground-truth target y. In the context of the MAVIC-T challenge, four translation tasks are defined: SAR→EO ($1 \rightarrow 1$ channel, 256×256), SAR→RGB ($1 \rightarrow 3$, 1024×1024), SAR→IR ($1 \rightarrow 1$, 1024×1024), and RGB→IR ($3 \rightarrow 1$, 1024×1024).

Evaluation Metric The MAVIC-T challenge evaluates submissions using a composite score derived from three complementary metrics: LPIPS (Learned Perceptual Image Patch Similarity, based on VGG-16) for perceptual fidelity, FID (Fréchet Inception Distance, based on InceptionV3) for distributional similarity, and L1 norm for pixel-level structural accuracy. These are combined into a task score:

$$S_{\text{task}} = \frac{\frac{2}{\pi}\arctan(\text{FID}) + \text{LPIPS} + \text{L1}}{3}, \quad (1)$$

where the $\arctan$ normalization maps FID into the $[0, 1]$ range to balance its contribution against the other two metrics. The overall score is the average of the four task scores, with a penalty of $+1$ added for each unattempted domain, encouraging participants to develop generalizable solutions across all modality pairs.

Denoising Diffusion Bridge Models Unlike standard diffusion models that learn to generate data from pure Gaussian noise, *Denoising Diffusion Bridge Models* (DDBM) define a stochastic bridge process that directly connects two data distributions—the source modality $\mathcal{X}$ and the target modality $\mathcal{Y}$. Given a training pair (x, y) where x is the source and y is the target, the forward bridge process constructs a sequence of noisy intermediate states z_t for $t \in [0, T]$ such that $z_0 = y$ and $z_T \approx x$. Under the variance-preserving (VP) formulation, the noisy sample at time t is:

$$z_t = a_t \cdot x + b_t \cdot y + \sigma_t \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (2)$$

where a_t, b_t, σ_t are functions of the VP schedule parameters (β_d, $\beta_{\min}$) and the log signal-to-noise ratio at time t. The reverse process, parameterized by a denoising neural network D_θ, learns to invert this corruption, thereby learning the conditional transport from $\mathcal{X}$ to $\mathcal{Y}$. This bridge formulation is inherently more suited for paired image translation than unconditional diffusion, as the source image directly constrains the generative trajectory.

4. Method

We present our solution for the 4th MAVIC-T challenge based on Denoising Diffusion Bridge Models (DDBM). Our approach adapts the DDBM framework to cross-modal remote sensing translation by (1) conditioning on the source image via channel concatenation, (2) employing a Karras-weighted bridge denoising objective with the option of augmenting it with a MAVIC-T-aware auxiliary loss, (3) optionally operating in a VAE latent space for high-resolution tasks, and (4) integrating representation alignment from pre-trained remote sensing encoders. Fig. 2 provides an overview of our pipeline.

4.1. Architecture

UNet Denoiser Our denoiser is built upon a UNet2DModel backbone with Adaptive Group Normalization (ADM-style) that accepts timestep embeddings.

Figure 2. Overview of our DDBM-based translation pipeline. The source image x is concatenated with the noisy sample z_t and fed to a UNet denoiser. During training, the bridge forward process constructs z_t from the target y and source x. Optional modules include a VAE encoder for latent-space modeling, a frozen MaRS encoder for representation alignment, and a MAVIC-T-aware loss branch. During inference, iterative Heun sampling transforms x into the translated output $\hat{y}$.

The network is instantiated with a base channel count of 128 and resolution-dependent channel multipliers—for example, $(1, 1, 2, 2, 4, 4)$ for 256×256 inputs—yielding block output channels of $(128, 128, 256, 256, 512, 512)$. Each resolution level uses 2 residual blocks. Self-attention is applied at spatial resolutions of 32×32, 16×16, and 8×8 via `AttnDownBlock2D` and `AttnUpBlock2D` modules, while other levels use plain `DownBlock2D`/`UpBlock2D` without attention.

Source Conditioning via Concatenation To condition on the source image x, we adopt the channel-concatenation strategy: the pre-conditioned noisy sample $c_{\text{in}} \cdot z_t$ and the source image x are concatenated along the channel dimension before being fed to the UNet. This doubles the input channels from C to $2C$, where C is the operating channel count of the model (e.g., $C = 1$ for SAR→EO, $C = 3$ for RGB→IR). The UNet output channels remain C, producing the raw denoised prediction.

Bridge Scalings Following DDBM, the raw model output is post-processed through bridge scalings $(c_{\text{skip}}, c_{\text{out}}, c_{\text{in}})$ derived from the VP log-SNR schedule. The denoised prediction is:

$$D_\theta(z_t, t, x) = c_{\text{out}}(t) \cdot F_\theta(c_{\text{in}}(t) \cdot z_t,\ t,\ x) + c_{\text{skip}}(t) \cdot z_t\ , \quad (3)$$

where F_θ is the raw UNet output and the scalings are computed from the VP parameters β_d and $\beta_{\min}$. The timestep is rescaled as $\hat{t} = 1000 \cdot 0.25 \cdot \log(\sigma + \varepsilon)$ (with $\varepsilon = 10^{-44}$ for numerical stability) before being passed to the model's sinusoidal embedding layer.

4.2. Training Objective

Bridge Denoising Loss The primary training objective is a weighted mean squared error between the denoised prediction $D_\theta(z_t, t, x)$ and the clean target y:

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{(x,y),\, t,\, \epsilon}\left[w(t) \cdot \|D_\theta(z_t, t, x) - y\|_2^2\right], \quad (4)$$

where $w(t) = A(t)/c_{\text{out}}(t)^2$ is the Karras bridge weighting derived from the combined variance $A(t) = a_t^2\sigma_{\text{data,end}}^2 + b_t^2\sigma_{\text{data}}^2 + c^2 c_t$ (with a_t, b_t, c_t being VP schedule coefficients), and z_t is the noisy sample obtained from the forward bridge process (Eq. 2). The diffusion time t (parameterized as sigma σ) is sampled from a Karras inverse-ρ distribution with $\rho = 7$: $\sigma = (\sigma_{\max}^{-1/\rho} + u \cdot (\sigma_{\min}^{-1/\rho} - \sigma_{\max}^{-1/\rho}))^\rho$, $u \sim \mathcal{U}(0, 1)$, which concentrates samples at lower noise levels where fine detail is learned.

MAVIC-T-Aware Auxiliary Loss Since the challenge evaluation directly uses LPIPS and L1 as two of the three scoring components, we explore augmenting the denoising objective with a differentiable *MAVIC-T-aware* loss that directly optimizes toward the evaluation metric. Specifically, we convert the denoised prediction to the $[0, 1]$ range and compute:

$$\mathcal{L}_{\text{mavic}} = \lambda_{\text{lpips}} \cdot \text{LPIPS}(\hat{y}, y) + \lambda_{\text{l1}} \cdot \|\hat{y} - y\|_1, \quad (5)$$

where LPIPS uses a frozen VGG-16 backbone. FID is a distribution-level metric and cannot serve as a per-sample loss, so it is excluded from the training criterion. The total loss becomes $\mathcal{L} = \mathcal{L}_{\text{denoise}} + \lambda_{\text{mavic}} \cdot \mathcal{L}_{\text{mavic}}$ with default $\lambda_{\text{mavic}} = 0.1$.

4.3. Representation Alignment

Inspired by the REPA (REPresentation Alignment) technique, we explore aligning the denoised predictions with features from frozen, pre-trained remote sensing encoders. The intuition is that modality-specific foundation models capture rich semantic representations that can serve as an auxiliary supervision signal, encouraging the translation model to preserve semantic content.

Encoder Selection For SAR-input tasks (SAR→EO, SAR→IR, SAR→RGB), we use a frozen MaRS-SAR encoder—a SwinV2-Base model (`swinv2_base_window8_256`) pre-trained on SAR data with a feature dimension of 1024. For the RGB→IR task, we use a frozen MaRS-RGB encoder with the same architecture but pre-trained on RGB imagery. Both encoders extract multi-scale feature maps, and we use global average pooling (GAP) on the deepest stage to obtain a d-dimensional feature vector.

Projection and Loss A trainable 3-layer MLP projector maps the model's output features to the encoder's embedding space:

$$h = \text{Linear} \circ \text{SiLU} \circ \text{Linear} \circ \text{SiLU} \circ \text{Linear}(f_{\text{model}}), \quad (6)$$

where the hidden dimension is $2d$ and the output dimension matches the encoder's feature dimension d. The alignment loss is the negative cosine similarity, averaged over the batch:

$$\mathcal{L}_{\text{align}} = -\frac{1}{B}\sum_{i=1}^{B} \frac{\bar{h}_i \cdot \bar{e}_i}{\|\bar{h}_i\| \cdot \|\bar{e}_i\|}, \quad (7)$$

where $\bar{h}_i = h_i/\|h_i\|$ and $\bar{e}_i = e_i/\|e_i\|$ are L2-normalized projected model features and frozen encoder features, respectively. The encoder features are detached (stop-gradient) so only the projector and UNet receive gradients. The total loss is augmented as $\mathcal{L} = \mathcal{L}_{\text{denoise}} + \lambda_{\text{align}} \cdot \mathcal{L}_{\text{align}}$ with default $\lambda_{\text{align}} = 1.0$.

4.4. Latent-Space Modeling

For the high-resolution tasks (1024×1024), directly operating in pixel space is computationally expensive. We explore an ablation where a pre-trained Variational Autoencoder (VAE) is used to encode target images into a compact latent space, and an additional *latent-space L2 loss* encourages the translation model's output to be consistent with the reference VAE's encoding. Specifically, let E_{vae} denote the frozen VAE encoder (an `AutoencoderKL` from a pre-trained checkpoint). The latent loss is:

$$\mathcal{L}_{\text{latent}} = \|E_{\text{vae}}(\hat{y}) - E_{\text{vae}}(y)\|_2^2, \quad (8)$$

where $E_{\text{vae}}(\cdot)$ returns the scaled mean of the VAE's posterior distribution (i.e., $\mu \cdot s$ with scaling factor s). For single-channel images, channels are replicated to 3 before encoding. The latent loss provides a compressed, semantically rich supervision signal that complements the pixel-level denoising objective, and is weighted by $\lambda_{\text{latent}} = 1.0$.

4.5. Inference

At inference time, we follow the DDBM sampling procedure using a second-order Heun integrator. Starting from the source image x, the scheduler discretizes the sigma schedule into N steps using the inverse-ρ Karras schedule from $\sigma_{\max}$ to $\sigma_{\min}$. At each step i, the denoiser produces a prediction $D_\theta(z_i, \sigma_i, x)$, which is used to compute the SDE drift. A Heun corrector step evaluates the model a second time at σ_{i+1} to refine the update, yielding approximately $2N$ total function evaluations. Optional stochastic churn with ratio $\gamma = 0.33$ injects controlled noise between steps to improve sample diversity. We use $N = 40$ steps by default and set guidance $w = 1.0$ (no classifier-free guidance). When an EMA model is available (with decay 0.9999), we use the EMA weights for inference.

5. Experiments

5.1. Dataset

We use the MAVIC-T 2025 dataset provided by the challenge organizers, which incorporates diverse geospatial data sources including UNICORN, USGS EROS High Resolution Orthoimagery (HRO), and the UMBRA open data program. The dataset consists of spatially aligned 200m×200m image stacks indexed by WGS-84 top-left coordinates, covering multiple geographic regions (UC Davis, Manhattan, Bingham, Centerfield, among others) across three sensing modalities: SAR, EO/RGB, and IR.

The four translation tasks differ in channel configuration and spatial resolution:

- **SAR→EO**: 1-channel → 1-channel at 256×256;
- **SAR→RGB**: 1-channel → 3-channel at 1024×1024;
- **SAR→IR**: 1-channel → 1-channel at 1024×1024;
- **RGB→IR**: 3-channel → 1-channel at 1024×1024.

For the three higher-resolution tasks, we supplement the original training data with crop-augmented variants to increase training diversity. We also apply a quality filter to exclude corrupted or misaligned samples identified during a data cleaning pass.

5.2. Implementation Details

Model Configuration Our DDBM UNet uses a base channel count of 128 with channel multipliers $(1, 1, 2, 2, 4, 4)$, 2 residual blocks per level, and self-attention at resolutions 32, 16, and 8. For the concat conditioning mode, the UNet input channels are doubled (e.g., $2 \times 1 = 2$ for SAR→EO, $2 \times 3 = 6$ for RGB→IR). The VP noise schedule uses $\sigma_{\min} = 0.002$, $\sigma_{\max} = 1.0$, $\sigma_{\text{data}} = 0.5$, $\beta_d = 2.0$, and $\beta_{\min} = 0.1$.

Training All models are trained using the Prodigy optimizer with an initial learning rate of 1.0 (Prodigy adapts the learning rate automatically) and a constant learning rate schedule with 500 warmup steps. We use bfloat16 mixed precision throughout. For SAR→EO, we use a batch size of 32; for the three 1024×1024 tasks, we use a batch size of 8. EMA with decay 0.9999 is applied with warmup. Gradient norms are clipped to 1.0. Training runs for 100 epochs, and we save checkpoints after every epoch, retaining the best-performing one. All experiments are conducted on NVIDIA GPUs using the Hugging Face Accelerate library for multi-GPU support.

Data Preprocessing All images are resized to the target resolution using bilinear interpolation and normalized to the $[0, 1]$ range (dividing by 255 for uint8 or 65535 for uint16 TIFF inputs). During training, images are further scaled to $[-1, 1]$. When the model operates in a channel space different from the data (e.g., 1-channel SAR expanded to 3 channels for the RGB→IR model), channels are replicated and truncated as needed. Random horizontal flipping is optionally applied as data augmentation, with identical transforms applied to both source and target images.

5.3. Baselines

We compare our DDBM-based approach against five competitive baselines spanning GAN-based, diffusion bridge, and unconditional diffusion paradigms:

- **CUT** (Contrastive Unpaired Translation): A GAN-based method using a ResNet-9 generator with instance normalization, a PatchGAN discriminator (70×70 receptive field), and a multi-layer patch-wise NCE loss ($\tau = 0.07$, 256 patches, layers 0/4/8/12/16). Single forward pass at inference.
- **BiBBDM** (Bidirectional Brownian Bridge Diffusion Model): A bidirectional diffusion bridge using a Brownian Bridge noise schedule (m_t linearly interpolated between endpoints, variance $\text{var}_t = 2 \cdot m_t \cdot (1 - m_t)$). Uses the "dlns" objective that jointly predicts two components with doubled output channels. Same UNet backbone as DDBM.
- **I2SB** (Image-to-Image Schrödinger Bridge): Learns the minimum KL-divergence path between source and target distributions using a symmetric beta schedule ($\beta_{\max} = 0.3$, interval $= 1000$). Predicts noise labels normalized by forward standard deviation. Uses 100 function evaluations (NFE) at inference.
- **DDIB** (Dual Diffusion Implicit Bridges): Trains *two independent unconditional* diffusion models, one per domain, and translates via deterministic DDIM encoding (source → latent) followed by DDIM decoding (latent → target) with 250 steps and $\eta = 0$. Uses a Gaussian diffusion process with linear beta schedule.
- **Pix2Pix-Turbo**: A latent diffusion approach based on Stable Diffusion Turbo with LoRA fine-tuning (rank 8 for UNet, rank 4 for VAE). Adds skip connections between the VAE encoder and decoder at four resolution scales. Performs single-step denoising at $t = 999$ with a frozen CLIP text encoder. Uses LPIPS ($\lambda = 5.0$) and L2 ($\lambda = 1.0$) losses.

All diffusion bridge baselines (BiBBDM, I2SB) use the same UNet backbone architecture as our DDBM model (128 base channels, attention at 32/16/8) for fair comparison. All methods share the same data loading pipeline, augmentation strategy, and Prodigy optimizer with identical hyperparameters where applicable. Representation alignment and MAVIC-T-aware losses are available as optional modules for all baselines.

5.4. Quantitative Results

Tab. 1 presents the quantitative comparison across the four MAVIC-T translation tasks. **[TODO: Fill in numbers af-**

Table 1. Quantitative comparison across the four MAVIC-T translation tasks. We report LPIPS (↓), FID (↓), L1 (↓), and the composite Task Score (↓). Best results are in **bold**, second best are underlined.

Method	SAR→EO				SAR→RGB				SAR→IR				RGB→IR				Overall
	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	
CUT	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
BiBBDM	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
I2SB	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
DDIB	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Pix2Pix-Turbo	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
DDBM (Ours)	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 2. Ablation study on the SAR→EO task. "Denoise" is the standard bridge denoising loss; "+MAVIC" adds the LPIPS+L1 auxiliary loss; "+REPA" adds representation alignment; "+Latent" adds the VAE latent-space L2 loss.

Configuration	LPIPS↓	FID↓	L1↓	Score↓
Denoise only (baseline)	–	–	–	
+ MAVIC loss (λ=0.1)	–	–	–	
+ REPA (λ=1.0)	–	–	–	
+ Latent loss (λ=1.0)	–	–	–	
+ MAVIC + REPA	–	–	–	
+ MAVIC + REPA + Latent (full)	–	–	–	

ter experiments complete.] Our DDBM-based method is compared against all five baselines under identical training conditions. The composite task score (Eq. 1) provides a single number that balances perceptual quality (LPIPS), distributional fidelity (FID), and pixel accuracy (L1). The overall score averages the four task scores.

5.5. Qualitative Results

Fig. 3 shows qualitative translation examples for representative samples across all four tasks. **[TODO: Add qualitative analysis after experiments.]** We observe that diffusion bridge methods (DDBM, BiBBDM, I2SB) tend to produce smoother outputs with better global structure than the single-pass CUT baseline, while CUT can sometimes produce sharper local textures due to its adversarial training. DDIB, which uses two separate unconditional models, may struggle with precise spatial alignment since it relies on a shared latent space assumption. Pix2Pix-Turbo, despite its single-step inference advantage, tends to hallucinate fine details due to the lossy VAE compression at high resolutions.

5.6. Ablation Study

We conduct ablation studies to evaluate the contribution of each component in our framework. All ablations use the DDBM backbone with the same training protocol unless otherwise noted.

Effect of MAVIC-T-Aware Loss We compare the standard denoising-only objective against the augmented objective that includes the differentiable MAVIC criterion (LPIPS + L1). Since two of the three evaluation metrics (LPIPS and L1) are directly optimized, we expect this auxiliary loss to improve the task score, particularly on the LPIPS and L1 components. **[TODO: Fill in ablation numbers.]** However, we note that the MAVIC loss is computed on the denoised prediction at a random noise level during training, which means the prediction quality varies; thus, the auxiliary loss provides a noisy but useful gradient signal.

Effect of Representation Alignment Adding the REPA alignment loss with a frozen MaRS encoder provides an auxiliary semantic supervision signal. For SAR-input tasks, the MaRS-SAR encoder (SwinV2-Base, 1024-dim features) extracts domain-specific features from the source image, while the trainable projector maps the model's output into the same embedding space. We hypothesize that this alignment encourages the model to preserve high-level semantic content (e.g., building layouts, road structures) during translation. **[TODO: Fill in ablation numbers.]**

Effect of Latent-Space Modeling The latent-space L2 loss provides supervision through a frozen pre-trained VAE encoder. This encourages the model's output to be consistent with the VAE's learned representation, which captures mid-level visual features. The VAE encoder projects images from pixel space ($H \times W$) to a compact latent space ($H/8 \times W/8$), providing an efficient compressed supervision signal. For single-channel images, channels are automatically replicated to 3 before VAE encoding. **[TODO: Fill in ablation numbers.]**

6. Conclusion

In this paper, we presented our solution for the 4th Multi-modal Aerial View Image Challenge – Translation (MAVIC-T), based on Denoising Diffusion Bridge Models (DDBM). Our approach leverages the bridge diffusion formulation to directly learn conditional transport be-

Figure 3. Qualitative comparison across translation tasks. Each row shows a different input sample, with columns showing the source image, outputs from each baseline, our DDBM result, and the ground-truth target. Our method produces translations with better structural fidelity and fewer artifacts compared to the baselines. **[TODO: Add visual comparison grid.]**

tween source and target modalities, conditioned via channel concatenation with a UNet denoiser trained under Karras-weighted bridge scalings. We explored three complementary ablations: (1) a MAVIC-T-aware auxiliary loss that directly optimizes toward the challenge evaluation metric (LPIPS + L1), (2) representation alignment using frozen MaRS encoders from the remote sensing domain, and (3) latent-space modeling via a pre-trained VAE for compressed supervision. We systematically compared DDBM against five competitive baselines—CUT, BiBBDM, I2SB, DDIB, and Pix2Pix-Turbo—under controlled experimental conditions across all four translation tasks (SAR→EO, SAR→RGB, SAR→IR, RGB→IR). **[TODO: Add summary of key findings and final ranking.]**

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Supplementary Material

7. Additional Implementation Details

Optimizer Details We use the Prodigy optimizer for all experiments. Prodigy is an adaptive learning rate optimizer that automatically tunes the step size, which removes the need for extensive learning rate search. We set the nominal learning rate to 1.0 (as recommended for Prodigy) and use a constant schedule with 500 warmup steps. Gradient norms are clipped to 1.0, and we use bfloat16 mixed precision via the Hugging Face Accelerate library. For the CUT baseline, we use Adam with $\beta_1 = 0.5, \beta_2 = 0.999$ and separate optimizers for the generator and discriminator, following the CUT convention.

EMA Configuration We apply Exponential Moving Average (EMA) to the model parameters with a decay rate of 0.9999 and warmup enabled. The EMA shadow parameters are saved alongside the model checkpoint. At inference time, we use the EMA weights, which generally produce smoother and more stable outputs than the raw model weights.

Checkpoint Management Models are saved in the Hugging Face Diffusers format, which stores the UNet weights as safetensors, the scheduler configuration as JSON, and a `model_index.json` that records the pipeline class name (`DDBMPipeline`). This allows one-line model loading via `Pipeline.from_pretrained(path).to('cuda')` for inference. Checkpoints are optionally pushed to the Hugging Face Hub for reproducibility.

Multi-GPU Training Training is distributed across multiple GPUs using Hugging Face Accelerate. The process group timeout is set to 7200 seconds to accommodate potential synchronization delays. Gradient accumulation steps can be configured per-task, though we use 1 by default. Launch is via `accelerate launch -m src.ddbm_baseline.train_{task}`.

8. Baseline Architecture Details

CUT Architecture The CUT generator uses a ResNet-9 architecture with 64 base filters (`ngf=64`), instance normalization, and no dropout. The encoder consists of a 7×7 convolution followed by two stride-2 downsampling convolutions, then 9 residual blocks, and finally two transposed convolutions for upsampling. The discriminator is a PatchGAN with 3 convolutional layers (`n_layers_D=3`) and 64 base filters. The NCE feature network (`PatchSampleMLP`) projects features from 5 intermediate encoder layers (indices 0, 4, 8, 12, 16) to a 256-dimensional embedding space using per-layer 2-layer MLPs. The GAN loss uses LSGAN (least-squares) formulation. The total generator loss is $\mathcal{L}_G = \lambda_{\text{GAN}} \cdot \mathcal{L}_{\text{GAN}} + \lambda_{\text{NCE}} \cdot \mathcal{L}_{\text{NCE}}$ with $\lambda_{\text{GAN}} = \lambda_{\text{NCE}} = 1.0$. An identity NCE loss is also included, and the NCE terms are averaged as $\mathcal{L}_{\text{NCE, both}} = 0.5 \cdot (\mathcal{L}_{\text{NCE}} + \mathcal{L}_{\text{NCE, idt}})$.

BiBBDM Architecture BiBBDM uses the same UNet2DModel backbone as DDBM with 128 base channels and the concat conditioning mode. The key difference is the noise schedule: BiBBDM uses a Brownian Bridge schedule where the mean m_t linearly interpolates from $m_0 = 0.001$ to $m_T = 0.999$, and the variance is $\text{var}_t = 2 \cdot s \cdot m_t \cdot (1 - m_t)$ with scale $s = 2.0$. The default "dlns" objective predicts both the endpoint reconstruction and noise jointly, requiring the output channels to be doubled to $2C$. The loss is a combination of objective reconstruction loss (L1 by default) and optional endpoint losses weighted by separate λ terms. Stochasticity is controlled by $\eta = 1.0$.

I2SB Architecture I2SB also uses the same UNet backbone with concat conditioning. It differs in its noise schedule: a symmetric linear beta schedule where $\beta_t = (\sqrt{\beta_{\text{start}}} + t \cdot (\sqrt{\beta_{\text{end}}} - \sqrt{\beta_{\text{start}}}))^2$ for the first half of the interval, mirrored for the second half, with $\beta_{\max} = 0.3$. This creates a symmetric noise process where noise increases from both endpoints and peaks at the midpoint. The training objective is noise prediction: given z_t, the model predicts the noise label $\ell = (z_t - y)/\sigma_{\text{fwd}}(t)$, normalized by the forward cumulative standard deviation. Loss is MSE between predicted and true noise. Inference uses 100 function evaluations (NFE).

DDIB Architecture DDIB trains two separate unconditional UNet models—one for the source domain and one for the target domain. Unlike the bridge methods, the UNets do not receive condition images; each operates independently with standard input channels (C instead of $2C$). If `learn_sigma=True`, the output channels are doubled to predict both mean and variance. Translation works by: (1) encoding the source image into a shared latent space via DDIM inversion with the source model, then (2) decoding from that latent using the target model with DDIM forward sampling. Both operations use 250 DDIM steps with $\eta = 0$ (deterministic). The noise schedule is a standard linear Gaussian diffusion with $\beta_{\text{start}} = 0.0001 \cdot s$ and $\beta_{\text{end}} = 0.02 \cdot s$ where $s = 1000/T$.

Pix2Pix-Turbo Architecture Pix2Pix-Turbo is based on Stable Diffusion Turbo, fine-tuned with LoRA adapters. The UNet (UNet2DConditionModel) uses LoRA rank 8, targeting 13 module types including attention layers (`to_q`, `to_k`, `to_v`, `to_out`), convolutions (`conv1`, `conv2`, `conv_shortcut`, `conv_out`), and feed-forward layers. The VAE (`AutoencoderKL`) uses LoRA rank 4. A key architectural modification is the addition of skip connections between the VAE encoder and decoder: four 1×1 convolution layers (`skip_conv_1` through `skip_conv_4`) map encoder intermediate features to the decoder, with a learnable blending factor γ (initialized to 1). This preserves fine spatial details from the source image. A frozen CLIP text encoder provides prompt embeddings (e.g., "convert SAR image to electro-optical image"), computed once and reused. Inference performs a single denoising step at $t = 999$.

9. Dataset Processing Details

Data Sources The MAVIC-T dataset is a composite of three geospatial data sources: (1) UNICORN, (2) USGS EROS High Resolution Orthoimagery (HRO) with submeter resolution, and (3) the UMBRA open data program for satellite SAR. Each source undergoes preprocessing to ensure georectification and alignment, stored as GeoTIFFs with spatial referencing. A standardized chipping process divides the preprocessed data into spatially aligned 200m×200m image stacks indexed by their WGS-84 top-left coordinates. The training data spans multiple geographic regions including UC Davis, Manhattan, Bingham, and Centerfield.

Crop Augmentation For the three 1024×1024 tasks (SAR→RGB, SAR→IR, RGB→IR), we create crop-augmented training variants by randomly cropping and resizing regions from the original training images. These augmented splits (e.g., `sar2rgb_crop_aug`) are concatenated with the original training set when the `use_augmented` flag is enabled, effectively increasing the training data volume and spatial diversity.

Quality Filtering We apply a quality filter to exclude corrupted, misaligned, or low-quality training samples. A manifest file (`bad_samples.txt`) lists the paths of problematic images, which are skipped during data loading. This filtering is applied uniformly across all baselines and ablation experiments.

10. Per-Task Configuration Summary

Table 3. Per-task DDBM configuration summary.

Parameter	SAR→EO	SAR→RGB	SAR→IR	RG
Source channels	1	1	1	
Target channels	1	3	1	
Model channels	1	3	1	
Resolution	256	1024	1024	1
Batch size	32	8	8	
Crop augmentation	No	Yes	Yes	
REPA encoder	MaRS-SAR	MaRS-SAR	MaRS-SAR	MaR
Inference steps	40	40	40	

Tab. 3 summarizes the per-task configuration for our DDBM method. All tasks share the same UNet backbone architecture (128 base channels, channel multipliers auto-detected from resolution), Prodigy optimizer (lr=1.0), EMA decay (0.9999), VP noise schedule ($\sigma_{\min} = 0.002$, $\sigma_{\max} = 1.0$, $\beta_d = 2.0$, $\beta_{\min} = 0.1$), and training duration (100 epochs). The primary differences are the operating channel count, spatial resolution, batch size, and the choice of representation alignment encoder (MaRS-SAR for SAR-input tasks, MaRS-RGB for RGB→IR).

11. Additional Ablation Results

Tab. 4 extends the ablation study from the main paper to all four translation tasks. **[TODO: Fill in extended ablation numbers.]** This allows us to assess whether the ablation components have consistent effects across different modality pairs and resolutions, or whether certain components are more beneficial for specific tasks.

Inference Speed Comparison Tab. 5 reports the inference time per image for each method. CUT and Pix2Pix-Turbo are the fastest due to their single-pass nature. Among the iterative methods, DDBM with 40 steps offers a favorable speed-quality trade-off, while DDIB requires 250 DDIM steps per direction (500 total), making it the slowest.

12. Additional Qualitative Results

Figs. 4 to 6 provide additional qualitative results for three of the four translation tasks. **[TODO: Add qualitative analysis.]** These visualizations complement the main paper's qualitative comparison by showing a wider variety of scenes and input conditions, including challenging cases with complex urban structures, varying illumination, and dense vegetation.

Table 4. Extended ablation study across all four tasks. Each row adds one component to the DDBM baseline.

Configuration	SAR→EO				SAR→RGB				SAR→IR				RGB→IR			
	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Score	LPIPS	FID	L1	Scor
Denoise only	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
+ MAVIC loss	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
+ REPA	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
+ Latent loss	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Full (all three)	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Figure 4. Additional qualitative results for the SAR→EO task. Left to right: source SAR image, CUT, BiBBDM, I2SB, DDIB, DDBM (Ours), ground truth. **[TODO: Add visual comparison grid.]**

Table 5. Inference speed comparison (seconds per image).

Method	Steps (NFE)	Time (s)
CUT	1	–
Pix2Pix-Turbo	1	–
DDBM (Ours)	40 (∼80 NFE)	–
BiBBDM	100	–
I2SB	100	–
DDIB	250×2	–

CVPR
#*****

Figure 5. Additional qualitative results for the SAR→RGB task. **[TODO: Add visual comparison grid.]**

Figure 6. Additional qualitative results for the RGB→IR task. **[TODO: Add visual comparison grid.]**